

Daily Review

Booklets 1 and 2

- 1 What is a system?
- 2 Automatic doors: what is the input?
- 3 Which energy does a marble have at the top of the run?
- 4 Write down the formula for power.
- 5 Name a real project from Booklet 1.

Parts working together
to do a job

A sensor detects a
person

Gravitational potential

$$P = E / t$$

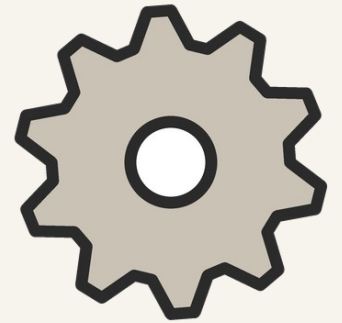
Edinburgh Trams or
Queensferry Crossing

TEST 1 REVISION • LESSON 1 OF 4

Engineers and What They Do

Who builds a tram — one engineer or four?

S3 Engineering Science



THE BIG QUESTION

Who builds a tram, or a bridge — one engineer, or four?



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PART 1

Engineers and What They Do

Name them, describe what they do, explain how they work.

By the end of this part...

- 1 I can name the 4 main types of engineering
- 2 I can describe what different engineers do
- 3 I can explain how engineers design, build, test and improve
- 4 I can explain my ideas clearly using what + why

SECTION A

Four Types of Engineer

Who designs the tunnel, and who designs the paint?

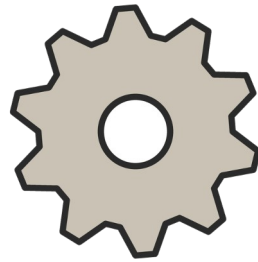
The Four Types of Engineer

What each one mainly works with — Booklet 1



Electrical

electricity,
wiring, power
and control



Mechanical

moving parts,
machines and
forces



Civil

buildings,
roads, bridges
and tunnels



Chemical

materials and
chemical
processes

One Project, Many Engineers — Edinburgh Trams

Electrical

power, lighting and displays

Mechanical

wheels, brakes and doors

Civil

tracks, stops and bridges

Chemical

paint and coatings that resist rust



Show Me — Which Engineer?

SHOW ME BOARDS

- 1 Which engineer designs a tunnel under a river?
Civil
- 2 ...works on the gears of a wind turbine?
Mechanical
- 3 ...designs the lighting for a football stadium?
Electrical
- 4 ...develops a paint that stops steel rusting?
Chemical
- 5 Name all four main types of engineering.
Electrical, mechanical, civil, chemical

Show Me — Which Engineer?

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Which engineer improves how drinking water is cleaned?
- 2 ...designs the brakes on a bike?
- 3 ...plans a new railway line?
- 4 ...designs the control system for traffic lights?
- 5 Why does a big project need more than one type of engineer?

Chemical

Mechanical

Civil

Electrical

Each works on a different part of it

SECTION B

Describe What Engineers Do

“Builds walls” — why does that score nothing?

Worked Example – Duns Park Flood Scheme

Describe one task a civil engineer would carry out. (1 mark)

What

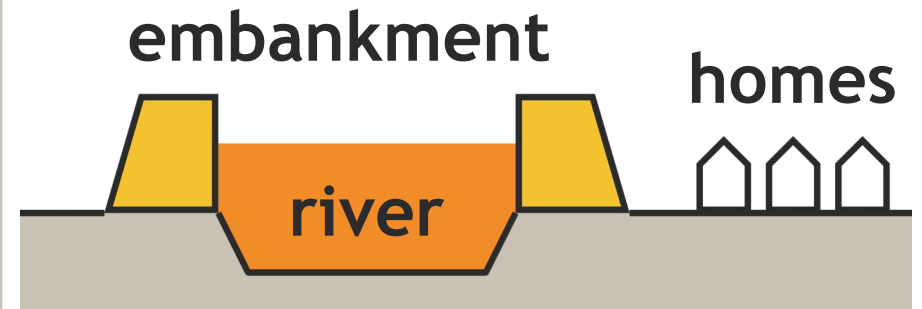
designs the flood embankments

Why

so nearby homes stay dry

Scores 0

“builds walls” – no detail, no why



Show Me — Queensferry Crossing

SHOW ME BOARDS

1 Civil engineer: one task, what + why

Designs the towers — so the deck stays up

2 Electrical engineer: one task, what + why

Lights the deck — so drivers can see at night

3 Mechanical engineer: one task, what + why

Designs moving joints — so the deck can expand

4 Chemical engineer: one task, what + why

Chooses the paint — so the steel won't rust

5 Which scores: A “fixes the lights” or B “lights the deck so drivers see”?

B — it says what AND why

Show Me — SpaceX Rocket

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Electrical engineer: one task, what + why
- 2 Mechanical engineer: one task, what + why
- 3 Chemical engineer: one task, what + why
- 4 Improve: “an electrical engineer does the wiring”
- 5 Why must every engineer's part be done properly?

Designs the controls —
so it can steer

Designs the engines — so
it can lift off

Develops the fuel — so it
burns to give thrust

Designs the wiring so the
controls get power

One failed part can stop
the whole system

SECTION C

Design, Build, Test, Improve

Why did your marble run need more than one go?

The Design Cycle – Round Until It Works

Prototype

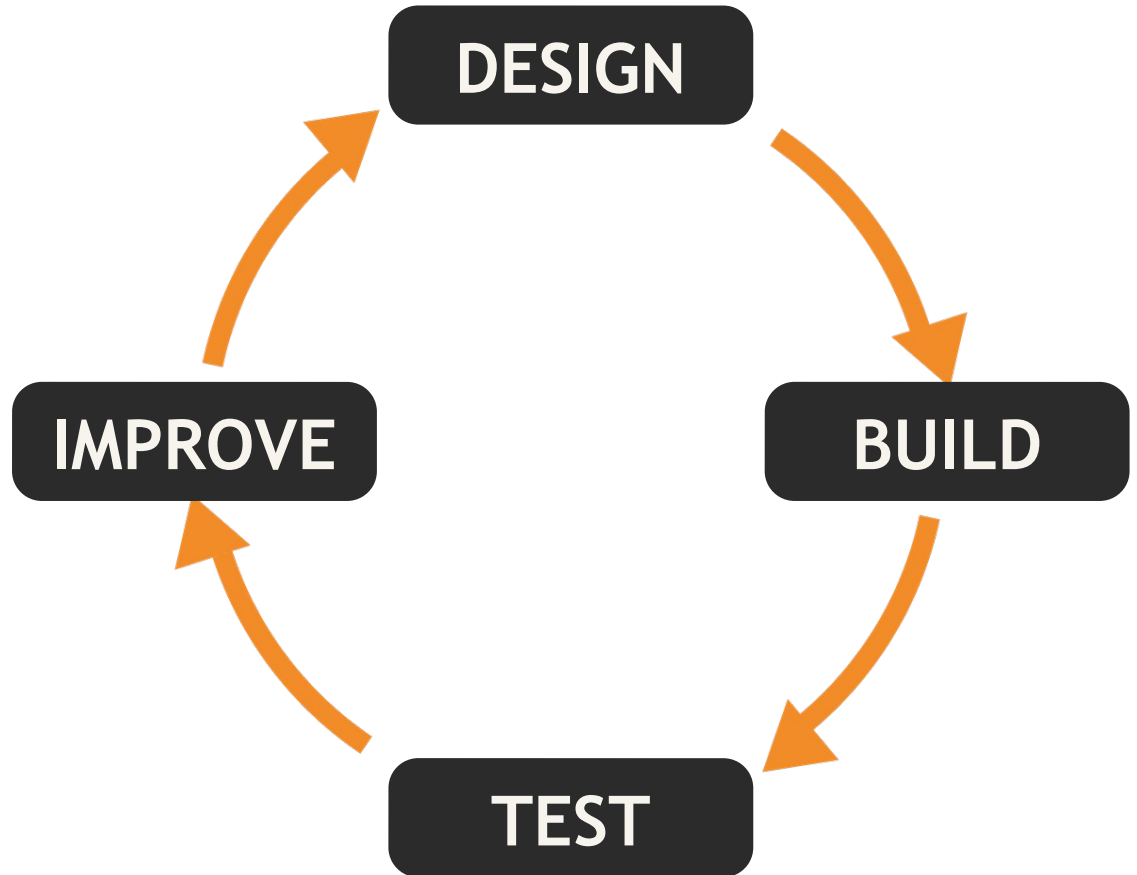
a first working version, built to be tested

Test

try it and measure it to find the faults

Improve

change the design, then test it again



Worked Example — Your Marble Run

Design

sketch the run on the board

Build

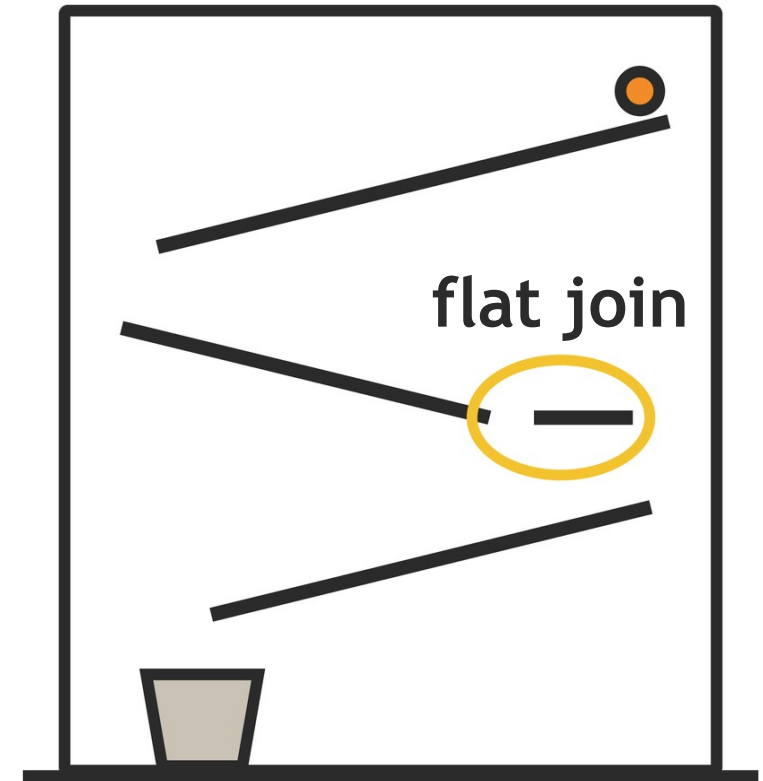
build the top section first

Test

the marble stops on a flat join

Improve

gentler join — then test again



Show Me – The Design Cycle

SHOW ME BOARDS

- 1 What is a prototype?
- 2 Put in order: test, improve, design, build.
- 3 A prototype drone crashes in testing. Which stage comes next?
- 4 Why test again after improving?
- 5 One thing you tested on your marble run – and what you improved.

A first working version,
built to test

Design, build, test,
improve

Improve

To check the change
fixed the fault

Your own answer

Show Me – The Design Cycle

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Which stage? “Sketches three ideas and picks one.”
- 2 Which stage? “Times three runs of the marble.”
- 3 Why is the cycle a loop, not a line?
- 4 A new school chair wobbles in testing. Improve it — what + why.
- 5 Why build one prototype before making 1000?

Design

Test

You improve, then test again

Add a brace — so it stays steady

To find faults before they cost a lot

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

What do engineers do?

- 1 fix broken machines
- 2 design, build, test and improve things
- 3 build things someone else has designed
- 4 do experiments in a lab

Answer 2

Engineers design, build, test and improve – a mechanic repairs.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

A company needs a coating that stops a bridge rusting. Which engineer?

- 1 civil
- 2 chemical
- 3 mechanical
- 4 electrical

Answer 2

The task decides the engineer: a coating is a material – chemical.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

Who designs the circuit that controls a lift's motor?

- 1 mechanical
- 2 civil
- 3 electrical
- 4 chemical

Answer 3

It moves, but a circuit is electrical. The motor's gears are mechanical.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

A prototype fails its test. What happens next?

- 1 the project is scrapped
- 2 it is sold anyway and fixed later
- 3 a new team builds it again the same way
- 4 the design is improved, then tested again

Answer 4

A failed test isn't the end – it tells you what to improve.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

“Describe one task a civil engineer would carry out on a new school.” Which answer gets the mark?

- 1 “builds it”
- 2 “civil engineering”
- 3 “works on the school”
- 4 “designs the foundations so it won't sink”

Answer 4

Only 4 says what the task is AND why it matters.

Mini Test 1

On your own, in silence, pen only.

8 minutes – 6 marks.

Then swap and mark from the next slides.

Mini Test 1 – Engineers and What They Do

Mini Test 1 — Marking

ANSWERS

Swap sheets. Tick or cross, then total out of 6.

Q1 (2 marks)

Electrical, mechanical, civil, chemical. All four = 2 marks. Two or three = 1 mark.

Q2 (1 + 1 marks)

(a) Electrical: designs the lighting, the wiring or the power for the pool pumps.

(b) Chemical: designs the water treatment that keeps the pool clean.

Mini Test 1 – Marking

ANSWERS

A task given to the wrong engineer scores 0.

Q3(a) (1 mark)

Design → build → test → improve

Q3(b) (1 mark)

Improve it: change the design or material (thicker corners, tougher plastic), then test it again to check it no longer cracks.

Scores 0

“improve it” on its own — say what you change and why.

Plenary

SHOW ME BOARDS

1

Name the four main types of engineering.

Electrical, mechanical,
civil, chemical

2

One task a chemical engineer does.

Develops fuels, plastics
or coatings

3

The four stages, in order.

Design, build, test,
improve

4

What + why: one job a civil engineer does.

Designs a bridge — so it
carries traffic safely

5

What is a prototype for?

Testing, to find the
faults

Plenary

SHOW ME BOARDS

EXTRA PRACTICE

1 Which engineer designs a car's engine?

Mechanical

2 Which engineer designs a dam?

Civil

3 Which stage finds the faults?

Test

4 Improve: “engineers make things”.

They design, build, test
and improve

5 Which I Can are you least sure of?

Your own answer

LOOKING AHEAD

Lesson 2 — Impacts of Engineering

Good and bad effects on people, money and the environment.